

Title

Randomised Comparison of Working Length Detection Methods in Root Canal Therapy: A Postoperative Discomfort Assessment Protocol

Abstract

Endodontic treatment carries risk of postoperative discomfort that may vary with procedural technique. Accurate determination of working length--the distance from tooth access to apical terminus--is fundamental to successful root canal completion. This study will compare postoperative pain patterns following root canal treatment when working length is established using one of several electronic detection systems versus traditional radiographic methods. A total of 125 patients with asymptomatic irreversible pulpitis affecting single-rooted teeth will be randomly allocated to five groups representing different working length determination approaches: four distinct electronic locator brands and conventional periapical radiography. A single operator will perform all treatments using standardised instrumentation and filling techniques in single-visit appointments. Pain intensity will be assessed via visual analogue rating at 6, 12, 24, and 48 hours post-completion. Medication consumption will be documented. This design will clarify whether working length detection methodology influences the magnitude or time-course of postoperative pain.